

MACHINE LEARNING METHODOLOGIES SUMMARY 2.

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From the resource shared to us to study I have this summary. It was a tutorial on exploratory data analysis (EDA) using different Python libraries to explore rollercoaster dataset. I got introduced to new libraries like numpy, matplotlib, and seaborn. Each of these has their different usage according to the tutor and with them we can do more with our dataset.

The video was categorized into different steps and I will elaborate a bit on each step so I can show my understanding of them. Other details about each step can be seen in the python script which I replicated from the tutor's codes. The tutor got his data from Kaggle where I also got mine ('coaster_db.csv') and after loading it, the EDA (Exploratory Data Analysis) started.

STEP 0: Imports & Reading the data.

Here we called all the Python libraries to be used and like I mentioned earlier they included, pandas, numpy, matplotlib and seaborn. Some settings were also made to matplotlib and pandas. Next the data was loaded using the 'pd.read_csv' function and we moved to the next step.

STEP 1: Data Understanding.

At this point, we want to know what our data is made up of so we looked at dataframe shape, head & tails, dtypes and describe. This step gave more information about the characteristics of the dataset where we got to know the number of columns and rows and other details.

STEP 2: Data Preparation.

Next, we looked into identifying duplicate columns, dropping irrelevant rows and columns, renaming columns and feature creation. We got to use the '.drop' function and also subsetting a dataset which we did by calling all the columns and commenting on the ones we don't want to use.

The subset was then defined as df and '.copy()' was added at the end so Python can understand that we are referencing the subset as a new data entirely and it doesn't relate to the original dataset.

STEP 3: Feature Understanding.

At this step, we delve into univariate analysis in which we looked at plotting feature distributions, histogram, KDE, Boxplot. The tutor showed us how to represent a single series, that is, a column with a histogram and other types of plot using matplotlib.

STEP 4: Feature Relationships.

Moving on into advanced plots, we started using seaborn which is not as limited as matplotlib to view the relationship between two or more series. This gave insights into how such series relate and can be used to answer some questions. We also checked for correlation where the tutor mentioned that correlation can only be used on columns with numerical values.

STEP 5: Ask a Question about the data.

At this last step, we attempted a question which was “ What are the locations with the fastest roller coasters?”. We answered the question and also visualized it with a horizontal barchart.